

ARB12: Addressing Geographic and Semantic Bias in Urban Street-View Segmentation for Arab Cities

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1. Motivation

- Street-view imagery supports walkability, comfort, and street-vitality analysis.
- Western-centric benchmarks show weaker transfer to Arab streetscapes.
- Arab streets contain active services and public-realm assets often absent from standard taxonomies.

CONSEQUENCE
Planning-relevant features collapse into generic classes, biasing urban indicators.

2. Research Gap

- Two barriers limit cross-region transfer:
- 1. Geographic shift:** Visual conditions differ across regions.
 - 2. Semantic mismatch:** Standard labels omit planning-relevant concepts.

3. Contributions

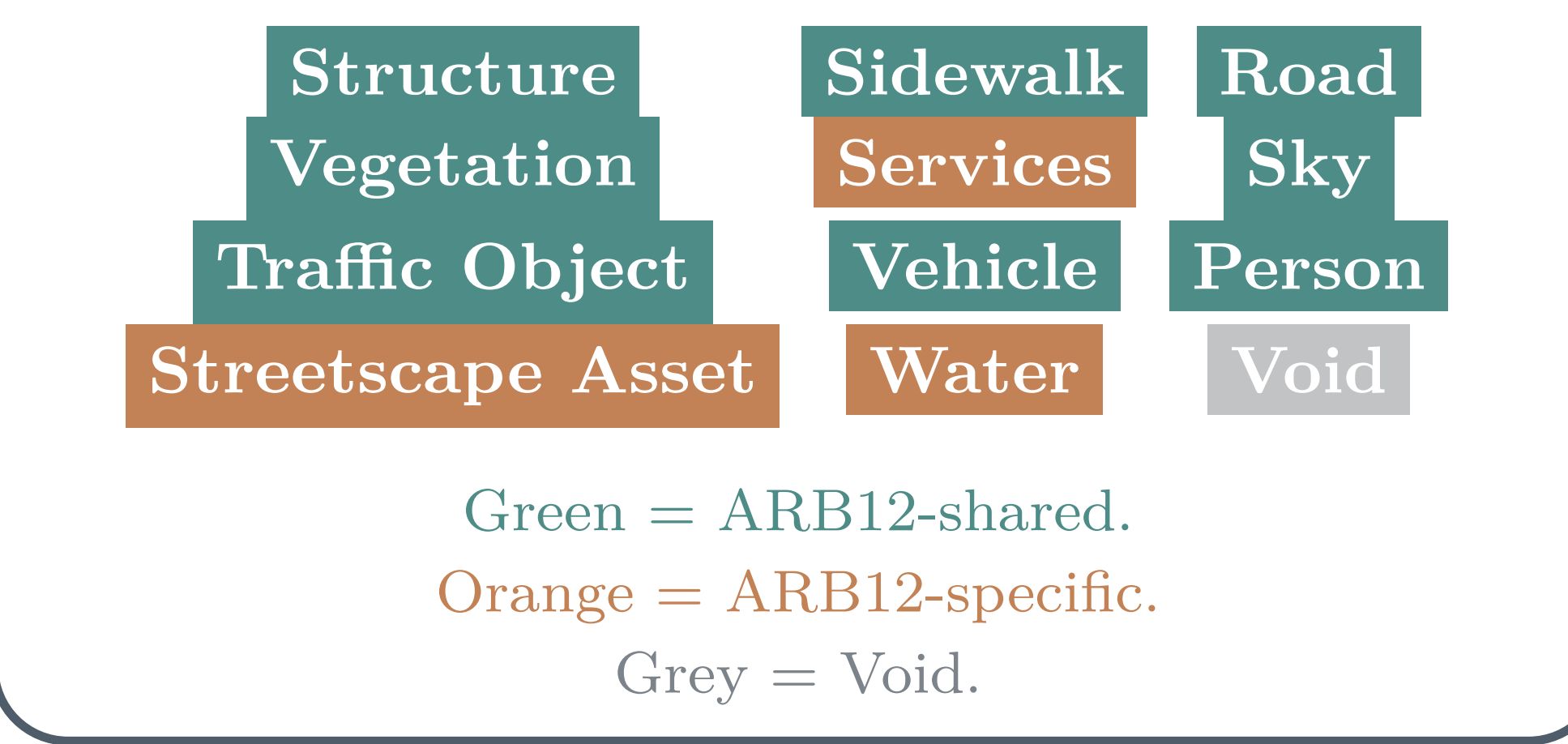
- 1. ARB12:** 12 region-aware classes for Arab streetscapes.
- 2. Eye-Level Rule:** Separates active Services from Structure.
- 3. Evaluation:** Distinguishes shared vs. ARB12-specific transfer performance.

4. ARB Saudi Dataset

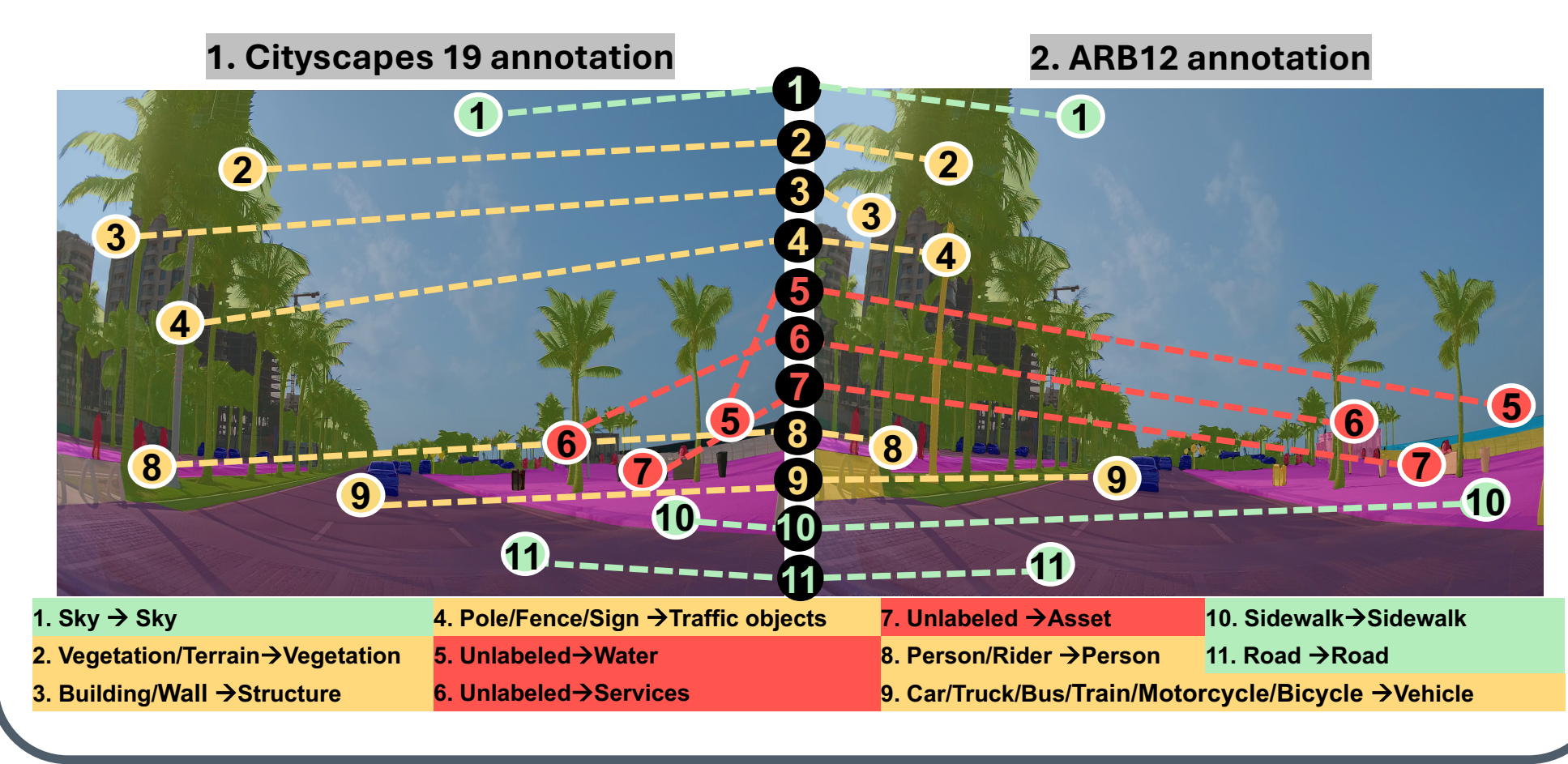
- **1,238** geo-referenced 4K videos
- **20.6 h** recording (**239 GB**)
- **10** Jeddah neighbourhoods
- Eye-level capture: **~1.6 m**, 140° FOV

ETHICS
Municipality-approved; identifiable data anonymised before use.

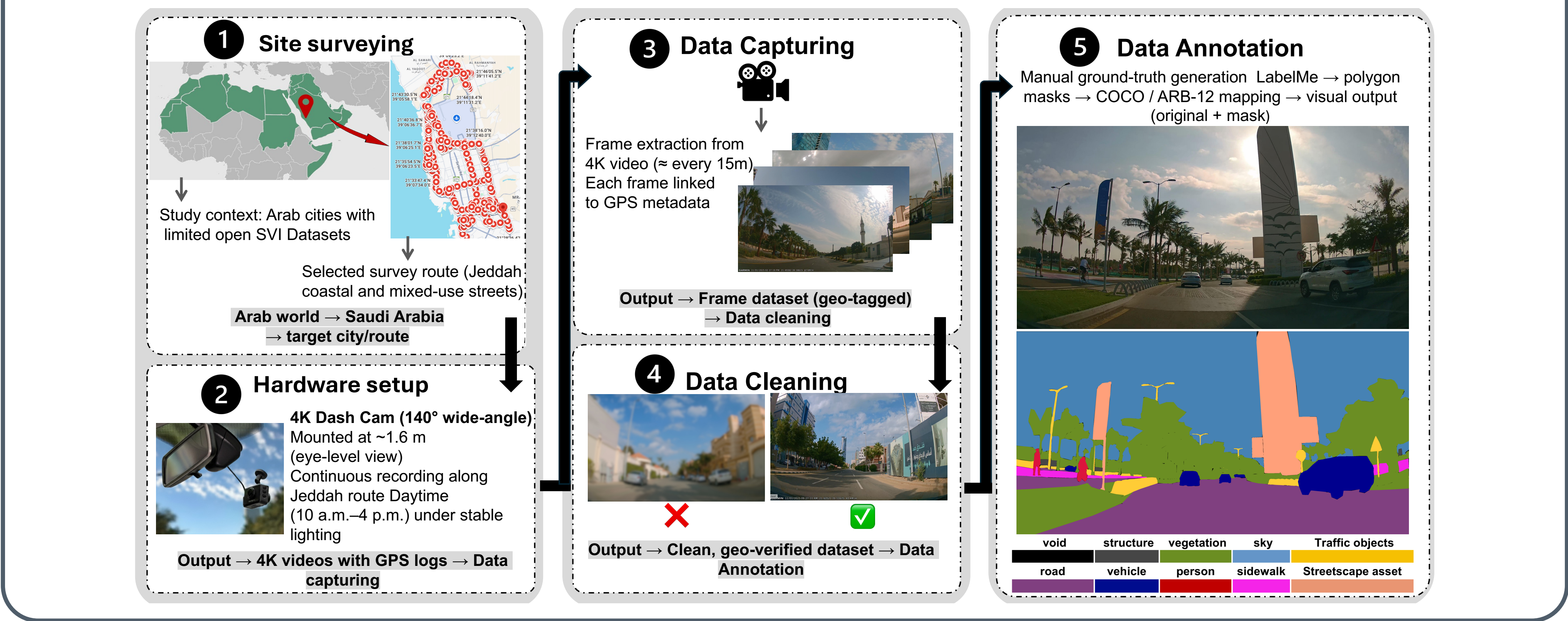
5. ARB12 Taxonomy



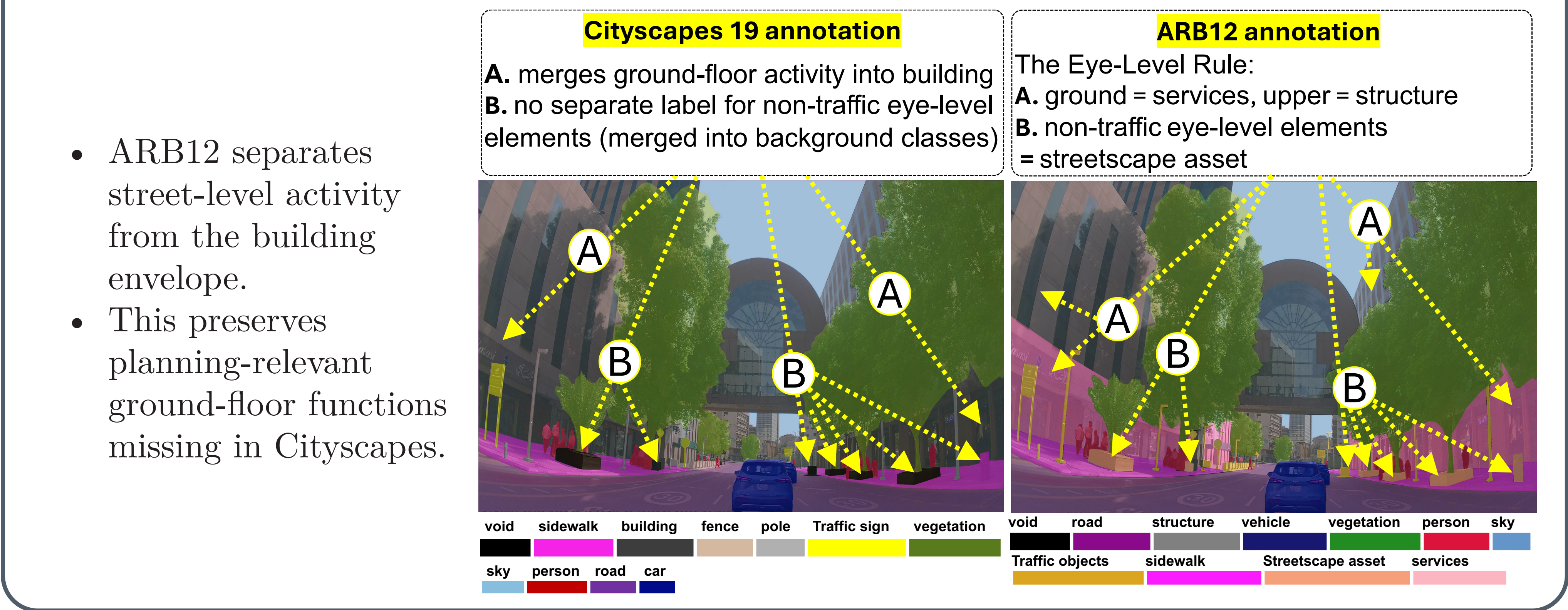
6. Mapping to ARB12



7. Acquisition Pipeline



8. Eye-Level Labelling



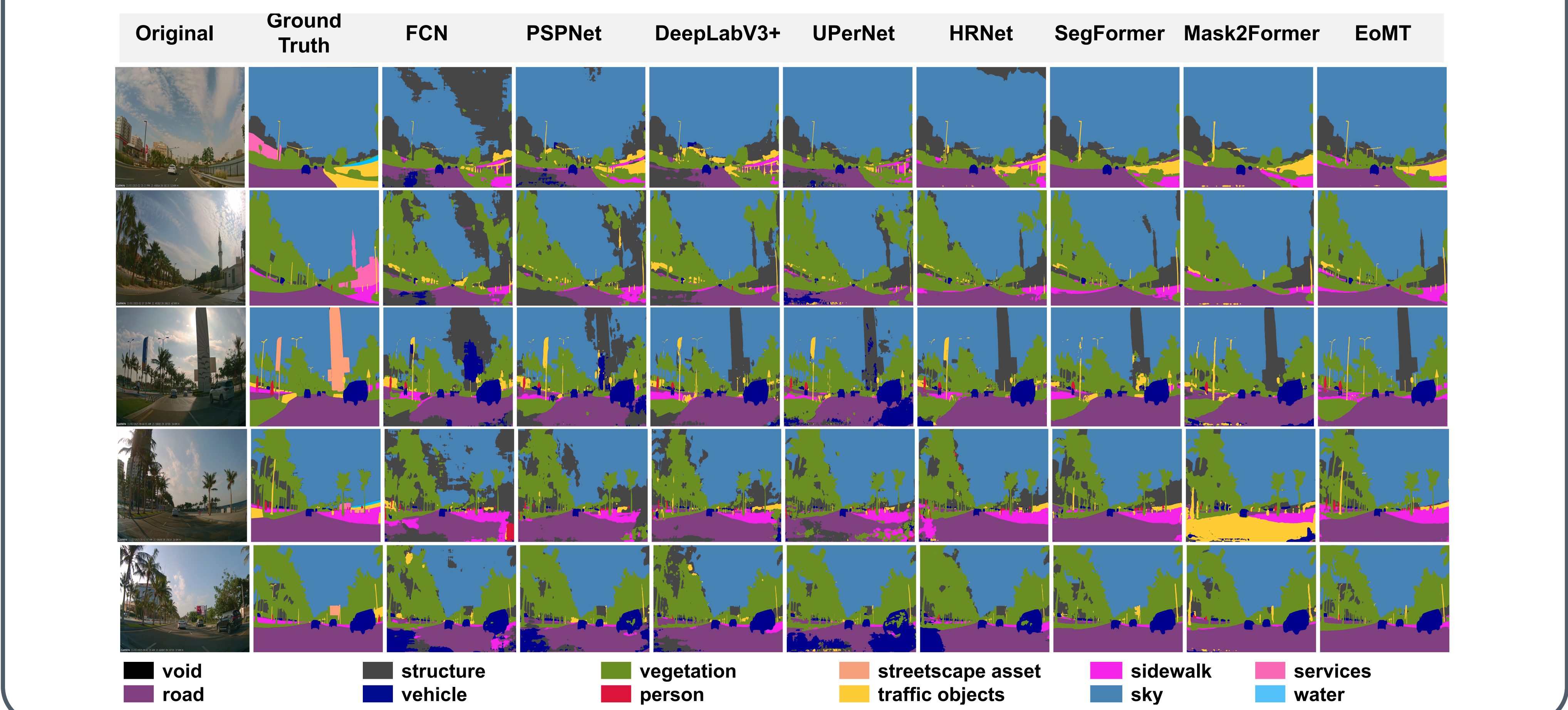
9. Class Quality

FCN	0.62	0.21	0.33	0.70	0.15	0.66	0.15	0.26
PSPNet	0.59	0.27	0.38	0.73	0.15	0.90	0.21	0.23
DeepLabV3+	0.66	0.32	0.51	0.70	0.16	0.93	0.23	0.26
UPerNet	0.61	0.36	0.35	0.71	0.10	0.94	0.21	0.16
HRNet	0.81	0.43	0.50	0.72	0.15	0.95	0.24	0.34
SegFormer	0.89	0.57	0.64	0.79	0.08	0.97	0.41	0.41
Mask2Former	0.75	0.49	0.43	0.75	0.07	0.96	0.20	0.34
EoMT	0.93	0.68	0.75	0.79	0.27	0.98	0.49	0.53
	road	structure	vehicle	vegetation	person	sky	Traffic objects	sidewalk

10. Boundary Quality

FCN	0.12	0.04	0.09	0.11	0.05	0.13	0.08	0.06
PSPNet	0.13	0.04	0.08	0.10	0.06	0.17	0.08	0.06
DeepLabV3+	0.14	0.06	0.12	0.11	0.05	0.24	0.10	0.08
UPerNet	0.12	0.06	0.09	0.11	0.03	0.24	0.10	0.05
HRNet	0.20	0.08	0.15	0.13	0.05	0.24	0.14	0.09
SegFormer	0.23	0.10	0.15	0.13	0.02	0.28	0.15	0.09
Mask2Former	0.13	0.06	0.06	0.09	0.01	0.20	0.04	0.04
EoMT	0.21	0.15	0.24	0.16	0.11	0.31	0.27	0.11
	road	structure	vehicle	vegetation	person	sky	Traffic objects	sidewalk

11. Qualitative Comparison



Conclusion

- Conclusion**
- Shared classes generalise relatively well.
 - ARB12-specific classes remain weak under zero-shot transfer.
 - Results reveal geographic and semantic bias.

Selected References

Selected References

- M. Cordts et al., “The Cityscapes Dataset for Semantic Urban Scene Understanding,” *Proc. IEEE CVPR*, 2016.
- S. N. Rai et al., “Munich to Dubai: How Far Is It for Semantic Segmentation?” *Proc. IEEE/CVF WACV*, 2020.
- T. Kerssies et al., “Your ViT is Secretly an Image Segmentation Model,” *Proc. IEEE/CVF CVPR*, 2025.
- B. Cheng et al., “Boundary IoU: Improving Object-Centric Image Segmentation Evaluation,” *Proc. IEEE/CVF CVPR*, 2021.

GitHub

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